

RES-5 a0-skeleton sensitivity: the 2PI gap-equation response is screened

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

The 2PI common-mode note reduced RES-5 to the a0-bound $|\Delta\Gamma_2^{\text{pd}}| \leq c a_0 \Delta F_{\text{margin}}$, with the constant c from the δG_*^{pd} sensitivity. The operator named the three steps: linearise the gap equation; bound $\delta G_*^{\text{pd}} = O(a_0)$; bound the skeleton sensitivity. This note carries out steps 1–2 and finds the response SCREENED (resolving the worry that $g > 1$ amplifies the feedback); step 3 reduces c to the SC-SCOPE skeleton self-energy norm. Scope: the screened response; anchor μ^2 .

2. Step 1 – linearise the 2PI gap equation

The gap equation $G^{-1} = K_0 + \lambda' \mathcal{P}^2 + \Sigma_2[G]$ linearised around the common $G_*^{\text{common}}(I)$ gives, with $\delta(G^{-1}) = -G^{-1} \delta G G^{-1}$,

$$\delta G_*^{\text{pd}} = -(1 + G_* K_{\text{BS}} G_*)^{-1} G_* \underbrace{\lambda'(\mathcal{P}^2 - \langle \mathcal{P}^2 \rangle)}_{\text{source}=O(a_0)} G_*, \quad K_{\text{BS}} = \frac{\delta \Sigma_2}{\delta G},$$

where K_{BS} is the Bethe–Salpeter (2PI) kernel and the source is the pattern-specific self-energy, $O(a_0)\hat{r}$.

3. Step 2 – the response is SCREENED (no strong-coupling amplification)

The response operator $(1 + G_* K_{\text{BS}} G_*)^{-1}$ is where a strong coupling could blow up. For the REPULSIVE density channel ($\lambda' = 3u_{\text{eff}} > 0$) the kernel is positive, $\|G_* K_{\text{BS}} G_*\| = g = \lambda' B_d = 1.03$, so

$$\|(1 + G_* K_{\text{BS}} G_*)^{-1}\| = \frac{1}{1 + g} = 0.49 \quad (\text{SCREENED, not } \frac{1}{1-g}).$$

The strong coupling $g > 1$ does NOT amplify δG_* : the sign is $1 + g$, not $1 - g$ – the SAME repulsive screening that made the susceptibility finite. Hence

$$\|\delta G_*^{\text{pd}}\| \leq \frac{1}{1 + g} a_0 \|G_*\| = 0.49 a_0 \|G_*\| = O(a_0),$$

ESTABLISHED including the self-consistent feedback. This resolves the ordered-BCC amplification worry: at strong coupling the feedback is screened, not divergent.

4. Step 3 – the skeleton sensitivity (the residual)

$$|\Delta\Gamma_2^{\text{pd}}| \leq \|\Sigma_2^{\text{pd}}\| \|\delta G_*^{\text{pd}}\| + \frac{1}{2}\|K_{\text{BS}}\| \|\delta G_*^{\text{pd}}\|^2,$$

with the second-order term $\frac{1}{2}g(0.49a_0)^2 = 0.001$ negligible. The linear term dominates: $c = \frac{\|\Sigma_2^{\text{pd}}\|}{\Delta F_{\text{margin}}} \times 0.49$, where $\|\Sigma_2^{\text{pd}}\|$ is the pattern-dependent skeleton self-energy – the SC-SCOPE SUN-SET scale (the $n = 3$ datum, $\sim 4\%$). Thus $c \sim 0.02$ and $ca_0 \sim 0.002 \ll 1$ plausibly. The rigorous $\|\Sigma_2^{\text{pd}}\|$ (the SC-SCOPE sunset self-energy, already computed at the difference level) is the sole remaining quantity.

5. Status

T3 PROOF SKETCH. Steps 1–2 ESTABLISHED: the gap-equation response is screened ($C_G = 0.49$), $\delta G_*^{\text{pd}} = O(a_0)$ including self-consistent feedback – the strong coupling does NOT amplify. The 2nd-order term is negligible. **OPEN (the residual):** the rigorous skeleton self-energy norm $\|\Sigma_2^{\text{pd}}\|$ (the constant c), which is the SC-SCOPE sunset scale – now to be assembled into the $ca_0 < 1$ certificate. No tier flip: B1-RH-ENUM T6 CONDITIONAL on {H-LAYER}. **B1 closure** $\iff$ this $\|\Sigma_2^{\text{pd}}\|$ bound, with everything else (common-mode mechanism, screened $O(a_0)$ response) in hand.

6. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “ $\|G_*K_{\text{BS}}G_*\| = g$ uses the same g – is the screening sign really guaranteed?” VALID-with-mitigation: the density-channel BS kernel inherits the sign of the dressed vertex $\lambda' > 0$ (repulsive), giving the $1 + g$ denominator; an attractive channel would give $1 - g$ (instability). The sign follows from $u_{\text{eff}} > 0$, robust. Off-diagonal/anomalous channels are a separate check (the exchange block, RES-1b).
- (β) “ $c \sim 0.02$ rests on the SC-SCOPE $\sim 4\%$.” VALID: c is reduced to the SC-SCOPE sunset self-energy norm, which is established at the difference level; the certificate must quote it rigorously, not the round 4%. But the STRUCTURE (screened $O(a_0)$ response $\times$ skeleton norm) is now fixed.
- (γ) “Does this close RES-5?” DISMISSED: steps 1–2 are established; step 3 (the skeleton norm) is the residual. B1 stays T6 on {H-LAYER}.

Quantitative sanity check (mandatory): *screening* – $1/(1 + g) = 0.49$ (finite, $g > 1$); *response* – $\delta G_*^{\text{pd}} = 0.49a_0\|G_*\| = O(a_0)$; *2nd order* – $\frac{1}{2}g(0.49a_0)^2 = 0.001$ negligible; *residual* – $c \sim \|\Sigma_2^{\text{pd}}\|/\Delta F_{\text{margin}} \times 0.49$, SC-SCOPE sunset scale; *margin* – $ca_0 \sim 0.002 \ll 1$ plausibly.

7. Result footer

Result ID: B1-RH-ENUM / RES-5 a0-skeleton sensitivity (screened response)

Precise statement: Linearising the 2PI gap equation, $\delta G_*^{\text{pd}} = -(1+G K_{\text{BS}} G)^{-1} G [\text{lam}'(P^2 - \langle P^2 \rangle)] G$. For the repulsive channel $|(1+G K_{\text{BS}} G)^{-1}| = 1/(1+g) = 0.49$ (SCREENED, not $1/(1-g)$), so $\delta G_*^{\text{pd}} = 0.49 a_0 \|G_*\| = O(a_0)$ -- strong coupling $g=1.03>1$ does NOT amplify the feedback

(resolves the ordered-BCC worry). 2nd-order term 0.001 negligible. Residual: $c = ||\Sigma_2^{\text{pd}}||/\Delta F_{\text{margin}} * 0.49$, $||\Sigma_2^{\text{pd}}|| = \text{SC-SCOPE sunset scale } (\sim 4\%)$; $c \sim 0.02$, $c a_0 \sim 0.002 \ll 1$ plausibly.

Scope: Steps 1-2 (screened a_0 -response); anchor μ^2 . The rigorous skeleton norm $||\Sigma_2^{\text{pd}}||$ (step 3) is OPEN.

Dependencies: res5-higherloop-commonmode-monotonicity (the 2PI structure); res5-2pi-resummation-strategy (the screening $1/(1+g)$); SC-SCOPE (sunset self-energy); Math424-AddA.

Evidence grade: ANALYTIC + EXECUTED (res5_a0_skeleton_sensitivity.py 4/4).

Reproduction command: python codes/vacuum/res5_a0_skeleton_sensitivity.py

Expected output: $C_G=1/(1+g)=0.49$; $\Delta G_{\text{pd}}=0.47$ (x a_0); 2nd-order 0.001; 4/4 PASS.

Falsification gate: An attractive density channel (would give $1/(1-g)$, $g>1$, amplification); or a skeleton self-energy $||\Sigma_2^{\text{pd}}||$ so large that $c a_0 \geq 1$.

Tier before / after: No tier flip. B1 T6 on {H-LAYER}. RES-5: the screened $O(a_0)$ gap-response established; residual reduced to the SC-SCOPE skeleton self-energy norm.

No-overclaim statement: This does NOT close RES-5. Steps 1-2 (screened, $O(a_0)$ response, no amplification) are established; the skeleton self-energy norm c (step 3) is the residual, = the SC-SCOPE sunset scale. The $c a_0 \sim 0.002$ estimate uses the round 4%; the certificate must quote $||\Sigma_2^{\text{pd}}||$ rigorously. B1 stays T6 on {H-LAYER}.

Next required action: Assemble the SC-SCOPE pattern-dependent sunset self-energy $||\Sigma_2^{\text{pd}}||$ into $c = ||\Sigma_2^{\text{pd}}||/\Delta F_{\text{margin}}*0.49$ and certify $c a_0 < 1$ -- the final RES-5 certificate.